The ratio of three quantities

Chaining two ratios into one, and sharing among three.

LESSON 47–49 | 3 × 40 MIN | MATEMATIKA 6, §11 | S7 12

LESSON CLOCK

25' | 30' | 30' | 25' | 10

Sharing among three	25 min
Chaining two ratios	30 min
From one quantity to all three	30 min
Mixtures	25 min
Homework	10 min

LEARNING OBJECTIVES

- Share a quantity among three in a given ratio.
- Combine $a : b$ and $b : c$ into $a : b : c$.
- Find all three quantities from one of them.
- Solve mixture problems with three ingredients.

TEXTBOOK

National	pp. 131–140
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.4

01 Explanation

Sharing among three

Nothing new: add all three parts, find one part, multiply out.

AMOUNT	RATIO	PARTS	SHARES
300	$1 : 2 : 3$	6	50, 100, 150
700	$1 : 2 : 4$	7	100, 200, 400
540	$2 : 3 : 4$	9	120, 180, 240
480	$3 : 5 : 8$	16	90, 150, 240

THE CHECK IS THE SAME

All three shares must add to the original amount. With three numbers to add there is more to go wrong, so the check matters more, not less.

Chaining two ratios

Given $A : B = 2 : 3$ and $B : C = 4 : 5$, the two Bs must be made equal before the ratios can be joined.

STEP	WORKING
the two ratios	$A : B = 2 : 3, B : C = 4 : 5$
make the Bs match — LCM of 3 and 4 is 12	$A : B = 8 : 12, B : C = 12 : 15$
join them	$A : B : C = 8 : 12 : 15$

$A : B$	$B : C$	$A : B : C$
$2 : 3$	$3 : 4$	$2 : 3 : 4$
$2 : 3$	$4 : 5$	$8 : 12 : 15$
$1 : 2$	$3 : 7$	$3 : 6 : 14$

ONLY THE SHARED QUANTITY IS MATCHED

Scaling $A : B$ by 4 and $B : C$ by 3 changes the numbers but not the ratios — that is exactly why it is allowed. Scaling only one of the two would break them.

From one quantity to all three

RATIO	GIVEN	ONE PART	ALL THREE
$2 : 3 : 5$	the first is 12	6	$12, 18, 30$
$1 : 4 : 6$	the second is 20	5	$5, 20, 30$
$3 : 4 : 5$	the third is 35	7	$21, 28, 35$
$2 : 3 : 4$	the total is 180	20	$40, 60, 80$

ALWAYS ONE PART FIRST

Whatever is given — a share, the total, or a difference — the first line of working is the value of one part. Every three-quantity problem then finishes in one more line.

Mixtures

PROBLEM	WORKING	ANSWER
concrete $1 : 2 : 4$ using 150 kg of cement	one part = 150	*1050 kg in all*
a salad $3 : 2 : 1$ of 600 g	one part = 100	*300, 200, 100 g*
brass $7 : 3$ copper to zinc in 500 g	one part = 50	*350 g copper*
a mix $2 : 3 : 5$ with 60 g of the largest part	one part = 12	*120 g in all*

READ WHICH PART THE GIVEN AMOUNT IS

“Using 150 kg of cement” names the 1; “with 60 g of the largest part” names the 5. The same number leads to very different totals depending on which part it is.

02 Worked examples

EXAMPLE 1 **Share 540 in the ratio $2 : 3 : 4$.**

1. $2 + 3 + 4 = 9$ parts.
2. $540 \div 9 = 60$ for one part.
3. 120, 180 and 240.
   Check: they total 540 ✓

ANSWER

120, 180, 240

EXAMPLE 2 **If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.**

1. The Bs are 3 and 4; their LCM is 12.
2. Scale the first by 4: $8 : 12$.
3. Scale the second by 3: $12 : 15$.
4. $A : B : C = 8 : 12 : 15$

ANSWER

$8 : 12 : 15$

EXAMPLE 3 **Concrete is $1 : 2 : 4$ cement, sand and gravel. A builder uses 150 kg of cement. What is the total mass?**

1. Cement is 1 part, so one part is 150 kg.
2. Total parts: 7.
3. $7 \cdot 150 = 1050$ kg.

ANSWER

1050 kg

03 Interactive model

Give the class two ratios sharing a middle quantity and ask them to join them without being shown how; most reach the LCM idea themselves.

Matching the shared quantity

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
3	*B in the first ratio*	$\times 4$
4	*B in the second*	$\times 3$

LOWEST COMMON DENOMINATOR

$$A : B : C = 8 : 12 : 15$$

The LCM of 3 and 4 is 12.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Three-part ratio	Uch hadli nisbat	Отношение трёх величин
To combine ratios	Nisbatlarni birlashtirish	Объединить отношения
Common term	Umumiy had	Общий член
Equivalent ratio	Teng nisbat	Равное отношение
Share	Ulush	Доля
Mixture	Aralashma	Смесь
Total parts	Jami ulushlar	Всего частей
Scaling	Miqyoslash	Масштабирование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

300 in $1 : 2 : 3$ gives:

- $50, 100, 150$
- $60, 100, 140$
- $100, 100, 100$
- $30, 60, 210$

To join $A : B$ and $B : C$ you make equal:

- the As
- the Bs
- the Cs
- the totals

$2 : 3$ and $4 : 5$ join to:

- $2 : 3 : 5$
- $8 : 12 : 15$
- $2 : 12 : 5$
- $6 : 12 : 20$

Ratio $2 : 3 : 5$ with the first 12 gives the third:

- 18
- 24
- 30
- 36

Concrete $1 : 2 : 4$ with 150 kg cement totals:

- 450 kg
- 750 kg
- 1050 kg
- 1200 kg

The first line of working is always:

- the total
- one part
- the ratio
- the check

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Share *300* in *1 : 2 : 3*

   ANSWER *50, 100, 150*

2. Share *700* in *1 : 2 : 4*

   ANSWER *100, 200, 400*

3. Share *540* in *2 : 3 : 4*

   ANSWER *120, 180, 240*

4. Share *480* in *3 : 5 : 8*

   ANSWER *90, 150, 240*

5. The parts of *2 : 3 : 5*

   ANSWER *10*

6. Ratio *2 : 3 : 5*, first *12*: one part

   ANSWER *6*

7. And all three

   ANSWER *12, 18, 30*

Medium · the standard question

7 PROBLEMS

1. $A : B = 2 : 3$, $B : C = 3 : 4$

   ANSWER $2 : 3 : 4$

2. $A : B = 2 : 3$, $B : C = 4 : 5$

   ANSWER $8 : 12 : 15$

3. $A : B = 1 : 2$, $B : C = 3 : 7$

   ANSWER $3 : 6 : 14$

4. Ratio $1 : 4 : 6$, second 20

   ANSWER $5, 20, 30$

5. Ratio $3 : 4 : 5$, third 35

   ANSWER $21, 28, 35$

6. Concrete $1 : 2 : 4$ with 150 kg cement

   ANSWER 1050 kg

7. A salad $3 : 2 : 1$ of 600 g

   ANSWER $300, 200, 100$ g

Hard · stretch and reason

7 PROBLEMS

1. Brass $7 : 3$ in 500 g: the copper

   ANSWER 350 g

2. A mix $2 : 3 : 5$ with 60 g of the largest part: the total

   ANSWER 120 g

3. $A : B = 3 : 4$, $B : C = 6 : 7$: find $A : B : C$

   ANSWER $9 : 12 : 14$

4. Share $2\ 400$ in $1 : 3 : 4$

   ANSWER $300, 900, 1200$

5. Ratio $2 : 3 : 4$, the difference between largest and smallest is 30

   ANSWER $30, 45, 60$

6. If $A : B = 2 : 5$ and $B : C = 10 : 3$, find $A : C$

   ANSWER $4 : 3$

7. Why must the shared quantity be matched before joining?

   ANSWER Otherwise the two ratios measure B in different units

07 Homework

Homework — 5 tasks

One part first; then every share is one multiplication away.

1. Share 720 in the ratio $2 : 3 : 7$.
2. If $A : B = 3 : 5$ and $B : C = 2 : 3$, find $A : B : C$.
3. A mixture is $1 : 3 : 6$ and uses 40 g of the first ingredient. Find the total mass.
4. Three shares are in the ratio $4 : 5 : 6$ and the largest is 42. Find the other two.
5. Concrete $1 : 2 : 4$ is needed to a total of $1\ 400$ kg. Find each ingredient.